

Mathematics for Data Science

Solution Sheet – Lecture 1

Vector Spaces, Subspaces, Linear Maps, Rank and Kernel

M1 – AI / Data Science

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Unless stated otherwise, all vector spaces are real vector spaces.

Part 1 – Course questions

Part 1(a) – Definitions

Question 1 (Euclidean space). *In machine learning, a data point is often represented by a vector $x = (x_1, \dots, x_d) \in \mathbb{R}^d$.*

(a) *Explain why $\mathbb{R}^d$ is a vector space. Which operations are involved?*

Answer. The set $\mathbb{R}^d$ is a vector space over $\mathbb{R}$ with the usual addition and scalar multiplication. If $x = (x_1, \dots, x_d)$ and $y = (y_1, \dots, y_d)$, then $x + y = (x_1 + y_1, \dots, x_d + y_d)$. If $\lambda \in \mathbb{R}$, then $\lambda x = (\lambda x_1, \dots, \lambda x_d)$. These operations satisfy the vector space axioms: closure, associativity, commutativity of addition, existence of the zero vector, existence of additive inverses, distributivity, compatibility of scalar multiplication, and $1x = x$.

(b) *Explain why the following set S is a subspace of $\mathbb{R}^d$:*

$$S = \left\{ x \in \mathbb{R}^d \mid \sum_{j=1}^d x_j = 0 \right\}.$$

Answer. First, $0 = (0, \dots, 0)$ belongs to S because $\sum_{j=1}^d 0 = 0$. Let $x, y \in S$. Then $\sum_{j=1}^d x_j = 0$ and $\sum_{j=1}^d y_j = 0$, hence $\sum_{j=1}^d (x_j + y_j) = 0 + 0 = 0$, so $x + y \in S$. Finally, if $\lambda \in \mathbb{R}$ and $x \in S$, then $\sum_{j=1}^d \lambda x_j = \lambda \sum_{j=1}^d x_j = \lambda \cdot 0 = 0$, so $\lambda x \in S$. Therefore S is a vector subspace of $\mathbb{R}^d$.

(c) *Give an example of a subset of $\mathbb{R}^d$ that is not a subspace, and explain which axiom fails.*

Answer. For example, the set $A = \{x \in \mathbb{R}^d \mid \sum_{j=1}^d x_j = 1\}$ is not a subspace. Indeed, $0 \notin A$, since $\sum_{j=1}^d 0 = 0 \neq 1$. Hence the zero vector axiom fails. Also, A is not stable under scalar multiplication: if $x \in A$, then $2x$ satisfies $\sum_{j=1}^d 2x_j = 2$, so $2x \notin A$.

Question 2 (Span). *In dictionary learning, one represents a data vector $x \in \mathbb{R}^d$ approximately as $x \approx c_1 d_1 + \dots + c_k d_k$, where $d_1, \dots, d_k \in \mathbb{R}^d$ are called dictionary atoms.*

(a) *Define the span of a family of vectors $d_1, \dots, d_k$.*

Answer. The span of $d_1, \dots, d_k$ is the set of all linear combinations of these vectors: $\text{span}(d_1, \dots, d_k) = \{c_1 d_1 + \dots + c_k d_k \mid c_1, \dots, c_k \in \mathbb{R}\}$.

- (b) *Explain why $\text{span}(d_1, \dots, d_k)$ is always a vector subspace.*

Answer. The zero vector belongs to the span by taking $c_1 = \dots = c_k = 0$. The sum of two linear combinations is again a linear combination: if $u = \sum_{j=1}^k c_j d_j$ and $v = \sum_{j=1}^k \alpha_j d_j$, then $u + v = \sum_{j=1}^k (c_j + \alpha_j) d_j$. Similarly, if $\lambda \in \mathbb{R}$, then $\lambda u = \sum_{j=1}^k (\lambda c_j) d_j$. Thus the span is stable under addition and scalar multiplication, so it is a subspace.

- (c) *What does it mean, geometrically, for a data point x to belong to $\text{span}(d_1, \dots, d_k)$?*

Answer. It means that x can be reconstructed exactly as a linear combination of the atoms $d_1, \dots, d_k$. Geometrically, x lies in the subspace generated by the dictionary atoms. In ML terms, the data point can be represented using these elementary patterns.

- (d) *What can go wrong if the dictionary atoms are linearly dependent?*

Answer. If the atoms are linearly dependent, at least one atom can be written as a linear combination of the others. This means that the dictionary contains redundant information. As a consequence, representations may fail to be unique: the same vector x may be written using different coefficient vectors $(c_1, \dots, c_k)$.

Question 3 (Basis, dimension, and embeddings). *In ML, a categorical variable with K categories is sometimes represented by one-hot vectors $e_1 = (1, 0, \dots, 0), \dots, e_K = (0, \dots, 0, 1)$ in $\mathbb{R}^K$.*

- (a) *Explain why $(e_1, \dots, e_K)$ is a basis of $\mathbb{R}^K$. What is the dimension of $\mathbb{R}^K$?*

Answer. The vectors $e_1, \dots, e_K$ span $\mathbb{R}^K$ because any vector $x = (x_1, \dots, x_K) \in \mathbb{R}^K$ can be written as $x = x_1 e_1 + \dots + x_K e_K$. They are also linearly independent: if $\alpha_1 e_1 + \dots + \alpha_K e_K = 0$, then the coordinates of this vector are $(\alpha_1, \dots, \alpha_K)$, hence all $\alpha_j = 0$. Therefore $(e_1, \dots, e_K)$ is a basis of $\mathbb{R}^K$, and $\dim(\mathbb{R}^K) = K$.

- (b) *Categories may also be represented by dense vectors in $\mathbb{R}^r$, with $r \ll K$. These vectors are called embeddings. Explain, in linear algebra terms, what changes when one replaces one-hot vectors in $\mathbb{R}^K$ by embeddings in $\mathbb{R}^r$.*

Answer. One-hot vectors represent categories in a K -dimensional space using the canonical basis of $\mathbb{R}^K$. Replacing them by embeddings in $\mathbb{R}^r$ means that categories are represented in a lower-dimensional vector space. Instead of assigning each category an orthogonal basis vector, we assign it a learned vector in $\mathbb{R}^r$. These embedding vectors may encode similarities between categories through their relative positions in $\mathbb{R}^r$.

Question 4 (Linear maps, kernel, range, rank). *Let $L : U \rightarrow V$ be a linear map.*

- (a) *Define $\ker(L)$, $\text{range}(L)$, and $\text{rank}(L)$.*

Answer. The kernel of L is $\ker(L) = \{u \in U \mid L(u) = 0_V\}$. The range, or image, of L is $\text{range}(L) = L(U) = \{v \in V \mid \exists u \in U, v = L(u)\}$. The rank of L is $\text{rank}(L) = \dim(\text{range}(L))$.

- (b) *State the rank-nullity theorem.*

Answer. If U is finite-dimensional and $L : U \rightarrow V$ is linear, then $\text{rank}(L) + \dim(\ker(L)) = \dim(U)$.

- (c) *In a neural network, a dense layer without bias and without activation has the form $x \mapsto Wx$. Explain why this is a linear map.*

Answer. Let $L(x) = Wx$. For any x, y and any scalars $\alpha, \beta \in \mathbb{R}$, we have $L(\alpha x + \beta y) = W(\alpha x + \beta y) = \alpha Wx + \beta Wy = \alpha L(x) + \beta L(y)$. Hence L is linear.

(d) Why is the map $x \mapsto Wx + b$ generally not linear?

Answer. A linear map must send 0 to 0. If $f(x) = Wx + b$, then $f(0) = b$. Therefore, if $b \neq 0$, $f(0) \neq 0$, so f is not linear. It is an affine map.

Part 1(b) – True/False questions

For each statement, decide whether it is true or false, and justify your answer.

T/F 1. Every subset of $\mathbb{R}^d$ containing the zero vector is a vector subspace.

Answer. False. Containing 0 is necessary but not sufficient. For example, in $\mathbb{R}$, the set $\{0, 1\}$ contains 0 but is not stable under addition, since $1 + 1 = 2 \notin \{0, 1\}$.

T/F 2. If $S \subseteq \mathbb{R}^d$ is closed under addition and scalar multiplication, and S is nonempty, then S is a vector subspace.

Answer. True. Since S is nonempty, take $x \in S$. By stability under scalar multiplication, $0x = 0 \in S$. Together with closure under addition and scalar multiplication, this proves that S is a subspace.

T/F 3. The set $A = \left\{x \in \mathbb{R}^d \mid \sum_{j=1}^d x_j = 1\right\}$ is a vector subspace of $\mathbb{R}^d$.

Answer. False. The zero vector does not belong to A , since $\sum_{j=1}^d 0 = 0 \neq 1$. This set is an affine hyperplane, not a vector subspace.

T/F 4. The span of a finite family of vectors is always a vector subspace.

Answer. True. The span is the set of all linear combinations of the family. It contains 0 and is stable under addition and scalar multiplication.

T/F 5. If $v_1, \dots, v_k$ are linearly independent vectors in $\mathbb{R}^d$, then $k \leq d$.

Answer. True. In a d -dimensional vector space, any linearly independent family has at most d vectors.

T/F 6. If $v_1, \dots, v_d$ span $\mathbb{R}^d$, then they form a basis of $\mathbb{R}^d$.

Answer. True. A spanning family of d vectors in a d -dimensional vector space is a basis. Equivalently, it must also be linearly independent.

T/F 7. If $L : \mathbb{R}^d \rightarrow \mathbb{R}^k$ is linear and $k < d$, then L cannot be injective.

Answer. True. Since $\text{rank}(L) \leq k < d$, rank-nullity gives $\dim(\ker(L)) = d - \text{rank}(L) > 0$. Therefore $\ker(L) \neq \{0\}$, so L is not injective.

T/F 8. If $\ker(L) = \{0\}$, then L is injective.

Answer. True. For a linear map, injectivity is equivalent to having a trivial kernel.

T/F 9. If $L : \mathbb{R}^d \rightarrow \mathbb{R}^k$ is linear and $\text{rank}(L) = k$, then L is surjective.

Answer. True. The range of L is a subspace of $\mathbb{R}^k$ of dimension k . Therefore $\text{range}(L) = \mathbb{R}^k$, so L is surjective.

T/F 10. In a data matrix, if one feature is an exact linear combination of other features, then the corresponding columns are linearly dependent.

Answer. True. Features are represented by columns. If one column is a linear combination of other columns, then the family of columns is linearly dependent.

T/F 11. A PCA-type representation of a data point by coordinates in a low-dimensional subspace is an example of representing data using linear combinations of basis directions.

Answer. True. In a PCA-type representation, a vector is approximated using coordinates along selected directions $u_1, \dots, u_k$. The reconstruction has the form $c_1u_1 + \dots + c_ku_k$, which is a linear combination of basis directions of a low-dimensional subspace.

Part 2 – Computational exercise

Exercise 1. Let's assume that a ML system represents online products using three types of features: $x = (t, e, i) \in \mathbb{R}^4 \times \mathbb{R}^3 \times \mathbb{R}^3$, where $t = (t_1, t_2, t_3, t_4) \in \mathbb{R}^4$ contains tabular numerical features, $e = (e_1, e_2, e_3) \in \mathbb{R}^3$ is a text embedding, and $i = (i_1, i_2, i_3) \in \mathbb{R}^3$ is an image embedding. The full representation space is $V = \mathbb{R}^4 \times \mathbb{R}^3 \times \mathbb{R}^3$. Assume that admissible product representations satisfy $t_2 = 100t_1$, $t_4 = t_1 + t_3$, $e_1 + e_2 + e_3 = 0$, and $i \in \text{span}(a, b)$, where $a = (1, 0, 1)$ and $b = (0, 1, 1)$. Let $S \subseteq V$ denote the set of all admissible representations.

1. What is the dimension of the vector space V ?

Answer. Since $V = \mathbb{R}^4 \times \mathbb{R}^3 \times \mathbb{R}^3$, we have $\dim(V) = 4 + 3 + 3 = 10$.

2. Show that S is a vector subspace of V .

Answer. The constraints defining S are homogeneous linear constraints, and the condition $i \in \text{span}(a, b)$ also defines a subspace of $\mathbb{R}^3$. More explicitly, if two admissible representations satisfy the constraints, then their sum also satisfies them. Similarly, multiplying an admissible representation by a scalar preserves all the constraints. The zero vector also satisfies all the constraints. Therefore S is a vector subspace of V .

3. Describe separately S_t , S_e , and S_i . Compute the dimensions of S_t , S_e , and S_i .

Answer. For S_t , the constraints are $t_2 = 100t_1$ and $t_4 = t_1 + t_3$. Thus t_1 and t_3 are free. If we write $t_1 = \alpha$ and $t_3 = \beta$, then $t = (\alpha, 100\alpha, \beta, \alpha + \beta)$. Hence $S_t = \text{span}((1, 100, 0, 1), (0, 0, 1, 1))$, and a basis of S_t is $((1, 100, 0, 1), (0, 0, 1, 1))$, so $\dim(S_t) = 2$.

For S_e , the constraint is $e_1 + e_2 + e_3 = 0$. If $e_1 = \alpha$ and $e_2 = \beta$, then $e_3 = -\alpha - \beta$, so $e = (\alpha, \beta, -\alpha - \beta)$. Hence $S_e = \text{span}((1, 0, -1), (0, 1, -1))$. A basis of S_e is $((1, 0, -1), (0, 1, -1))$, so $\dim(S_e) = 2$.

For S_i , we have $S_i = \text{span}(a, b)$ with $a = (1, 0, 1)$ and $b = (0, 1, 1)$. The vectors a and b are linearly independent, so $\dim(S_i) = 2$. Suppose $\alpha a + \beta b = 0$. Then $\alpha(1, 0, 1) + \beta(0, 1, 1) = (\alpha, \beta, \alpha + \beta) = (0, 0, 0)$. Thus $\alpha = 0$ and $\beta = 0$. Therefore a and b are linearly independent. A basis of S_i is $((1, 0, 1), (0, 1, 1))$.

NB: Since $S = S_t \times S_e \times S_i$, we get $\dim(S) = \dim(S_t) + \dim(S_e) + \dim(S_i) = 2 + 2 + 2 = 6$. A basis of S is obtained by combining the bases of S_t , S_e , and S_i .

Now, we consider a model that does not use the full representation $x = (t, e, i)$. Instead, it keeps only the following four quantities: $\Phi(x) = (t_1, t_3, e_1 - e_2, i_1 + i_2 + i_3)^\top \in \mathbb{R}^4$.

4. Show that $\Phi : S \rightarrow \mathbb{R}^4$ is a linear map.

Answer. Each coordinate of $\Phi(x)$ is a linear expression in the coordinates of x . Therefore, for $x, y \in S$ and $\lambda, \mu \in \mathbb{R}$, we have $\Phi(\lambda x + \mu y) = \lambda\Phi(x) + \mu\Phi(y)$. Hence Φ is linear.

5. Determine whether Φ is injective.

Answer. We will see that $\ker(\Phi)$ is not reduced to $\{0\}$, so Φ is not injective.

6. Determine $\ker(\Phi)$ and compute $\dim(\ker(\Phi))$.

Answer. Let $x = (t, e, i) \in S$. Write $t = \alpha(1, 100, 0, 1) + \beta(0, 0, 1, 1)$, $e = \gamma(1, 0, -1) + \delta(0, 1, -1)$, and $i = \mu(1, 0, 1) + \nu(0, 1, 1)$. Then $t_1 = \alpha$, $t_3 = \beta$, $e_1 - e_2 = \gamma - \delta$, and $i_1 + i_2 + i_3 = 2\mu + 2\nu$. Thus $\Phi(x) = 0$ iff $\alpha = 0$, $\beta = 0$, $\gamma = \delta$, and $\mu + \nu = 0$.

Therefore the kernel is

$$\ker(\Phi) = \{(0, \gamma(1, 1, -2), \mu(1, -1, 0)) \mid \gamma, \mu \in \mathbb{R}\},$$

where 0 denotes the zero vector in the tabular component $\mathbb{R}^4$. Hence $\dim(\ker(\Phi)) = 2$.

7. Use the rank-nullity theorem to compute $\text{rank}(\Phi)$.

Answer. Since $\Phi : S \rightarrow \mathbb{R}^4$ and $\dim(S) = 6$, the rank-nullity theorem gives $\text{rank}(\Phi) + \dim(\ker(\Phi)) = 6$. Since $\dim(\ker(\Phi)) = 2$, we obtain $\text{rank}(\Phi) = 4$.

8. Is Φ surjective onto $\mathbb{R}^4$?

Answer. Yes. Since $\text{rank}(\Phi) = 4$ and the codomain is $\mathbb{R}^4$, the range has dimension 4. Therefore $\text{range}(\Phi) = \mathbb{R}^4$, so Φ is surjective.

9. Interpret $\ker(\Phi)$ in terms of information lost by the feature extraction map Φ .

Answer. The kernel consists of admissible variations in the original representation that are completely invisible after feature extraction. Here, variations of the form $e = \gamma(1, 1, -2)$ and $i = \mu(1, -1, 0)$ are lost by Φ . In ML terms, two product representations that differ by an element of $\ker(\Phi)$ have exactly the same extracted features, so the downstream model cannot distinguish them.

The final prediction model is $g(x) = w^\top \Phi(x) + \beta$, where $w \in \mathbb{R}^4$ and $\beta \in \mathbb{R}$.

10. For fixed w , show that $x \mapsto w^\top \Phi(x)$ is linear on S .

Answer. The map Φ is linear, and the map $z \mapsto w^\top z$ is a linear map from $\mathbb{R}^4$ to $\mathbb{R}$. The composition of two linear maps is linear. Therefore $x \mapsto w^\top \Phi(x)$ is linear on S .

11. Show that g is generally not linear when $\beta \neq 0$.

Answer. If g were linear, we would have $g(0) = 0$. But $g(0) = w^\top \Phi(0) + \beta = \beta$. Thus if $\beta \neq 0$, then $g(0) \neq 0$, so g is not linear.

12. Explain why bias terms are useful in machine learning.

Answer. Bias terms allow the model to shift the decision boundary or prediction function. Without a bias, a linear model is constrained to pass through the origin in the appropriate coordinate system. With a bias, the model is affine, which is usually more flexible and more realistic for data.

Part 3 – A bit of ML

Context

In computer vision, a grayscale image with d pixels can be represented as a vector $x = (x_1, \dots, x_d) \in \mathbb{R}^d$, where x_j is the intensity of pixel j . A common objective is to replace the high-dimensional vector $x \in \mathbb{R}^d$ by a lower-dimensional vector $z \in \mathbb{R}^k$, $k \ll d$, while keeping as much useful information as possible.

Preliminary definitions / before we define everything in the next lectures :)

For $x, y \in \mathbb{R}^d$, define the dot product by $x^\top y = \sum_{j=1}^d x_j y_j$. Assume that the vectors $u_1, \dots, u_k$ are orthonormal, meaning that $u_i^\top u_j = 1$ if $i = j$ and $u_i^\top u_j = 0$ if $i \neq j$.

ML set-up

For any image vector $x \in \mathbb{R}^d$, define $E : \mathbb{R}^d \rightarrow \mathbb{R}^k$ by $E(x) = (u_1^\top x, \dots, u_k^\top x)^\top$. Define also $D : \mathbb{R}^k \rightarrow \mathbb{R}^d$ by $D(z) = z_1 u_1 + \dots + z_k u_k$. Finally, define the reconstruction of x by $\hat{x} = D(E(x))$.

Step 1 – Subspaces and dimension

1. Explain why the image vectors belong to the vector space $\mathbb{R}^d$.

Answer. A grayscale image with d pixels is described by d real numbers, one for each pixel intensity. Therefore it can be represented as a vector $x = (x_1, \dots, x_d) \in \mathbb{R}^d$. The vector space is $\mathbb{R}^d$.

2. Show that $S = \text{span}(u_1, \dots, u_k)$ is a vector subspace of $\mathbb{R}^d$.

Answer. By definition, S is the set of all linear combinations of $u_1, \dots, u_k$. A span is always a vector subspace: it contains 0, and it is stable under addition and scalar multiplication.

3. What is $\dim(S)$? Justify your answer.

Answer. The vectors $u_1, \dots, u_k$ are orthonormal. In particular, they are linearly independent. Since they also span S by definition, they form a basis of S . Therefore $\dim(S) = k$.

4. Explain geometrically why replacing $x \in \mathbb{R}^d$ by an element of S can be seen as dimensionality reduction.

Answer. The original vector x lives in the d -dimensional space $\mathbb{R}^d$. The subspace S has dimension k , with $k \ll d$. Replacing x by an element of S means approximating it using only k independent directions instead of d coordinates. This is a reduction of dimension.

Step 2 – Encoding as a linear map

5. Show that $E : \mathbb{R}^d \rightarrow \mathbb{R}^k$ is linear.

Answer. Let $x, y \in \mathbb{R}^d$ and $\alpha, \beta \in \mathbb{R}$. For each j , we have $u_j^\top (\alpha x + \beta y) = \alpha u_j^\top x + \beta u_j^\top y$. Therefore $E(\alpha x + \beta y) = \alpha E(x) + \beta E(y)$, so E is linear.

6. What is the range of E ? Is E surjective?

Answer. The range of E is $\mathbb{R}^k$. Indeed, for any $z = (z_1, \dots, z_k) \in \mathbb{R}^k$, take $x = z_1 u_1 + \dots + z_k u_k$. Then, using orthonormality, $E(x) = z$. Hence E is surjective.

7. Use the rank-nullity theorem to compute $\dim(\ker(E))$.

Answer. Since $E : \mathbb{R}^d \rightarrow \mathbb{R}^k$ is surjective, $\text{rank}(E) = k$. By rank-nullity, $\text{rank}(E) + \dim(\ker(E)) = d$. Therefore $\dim(\ker(E)) = d - k$.

8. Describe $\ker(E)$ using the vectors $u_1, \dots, u_k$.

Answer. We have $x \in \ker(E)$ iff $E(x) = 0$, which means $u_1^\top x = 0, \dots, u_k^\top x = 0$. Hence $\ker(E) = \{x \in \mathbb{R}^d \mid u_j^\top x = 0 \text{ for all } j = 1, \dots, k\}$. This is the set of vectors orthogonal to all the directions $u_1, \dots, u_k$.

9. Give an interpretation for $\ker(E)$.

Answer. The kernel is the set of directions that are completely ignored by the encoder. If $h \in \ker(E)$, then $E(x + h) = E(x)$ for every x . Thus changing an image vector by an element of $\ker(E)$ does not change its embedding.

Step 3 – Decoding as a linear map

10. Show that $D : \mathbb{R}^k \rightarrow \mathbb{R}^d$ is linear.

Answer. Let $z, z' \in \mathbb{R}^k$ and $\alpha, \beta \in \mathbb{R}$. Then $D(\alpha z + \beta z') = \sum_{j=1}^k (\alpha z_j + \beta z'_j) u_j = \alpha D(z) + \beta D(z')$. Therefore D is linear.

11. What is $\text{range}(D)$?

Answer. By definition, $D(z) = z_1 u_1 + \cdots + z_k u_k$. Therefore $\text{range}(D) = \text{span}(u_1, \dots, u_k) = S$.

12. Is D injective? Justify your answer.

Answer. Yes. If $D(z) = 0$, then $z_1 u_1 + \cdots + z_k u_k = 0$. Since $u_1, \dots, u_k$ are linearly independent, all coefficients z_j must be zero. Thus $z = 0$, so $\ker(D) = \{0\}$ and D is injective.

13. Is D surjective onto $\mathbb{R}^d$ when $k < d$? Justify your answer.

Answer. No. The range of D is S , which has dimension k . Since $k < d$, S is a strict subspace of $\mathbb{R}^d$. Therefore D is not surjective onto $\mathbb{R}^d$.

Step 4 – Reconstruction

14. Write explicitly $\hat{x} = D(E(x))$.

Answer. Since $E(x) = (u_1^\top x, \dots, u_k^\top x)$, applying D gives $\hat{x} = D(E(x)) = (u_1^\top x) u_1 + \cdots + (u_k^\top x) u_k$.

15. Show that $\hat{x} = \sum_{j=1}^k (u_j^\top x) u_j$.

Answer. This follows directly from the definition of D . If $z = E(x)$, then $z_j = u_j^\top x$. Therefore $D(z) = \sum_{j=1}^k z_j u_j = \sum_{j=1}^k (u_j^\top x) u_j$.

16. Show that $\hat{x} \in S$ for every $x \in \mathbb{R}^d$.

Answer. The vector $\hat{x}$ is a linear combination of $u_1, \dots, u_k$. Therefore $\hat{x} \in \text{span}(u_1, \dots, u_k) = S$.

17. Suppose that $x \in S$. Show that $\hat{x} = x$.

Answer. Since $x \in S$, there exist coefficients $c_1, \dots, c_k$ such that $x = c_1 u_1 + \cdots + c_k u_k$. For each j , orthonormality gives $u_j^\top x = c_j$. Hence $\hat{x} = \sum_{j=1}^k (u_j^\top x) u_j = \sum_{j=1}^k c_j u_j = x$.

18. Explain why $\hat{x}$ can be interpreted as the part of x that is kept by the low-dimensional representation.

Answer. The representation $E(x)$ stores only the coordinates of x along the directions $u_1, \dots, u_k$. The reconstruction $\hat{x}$ is the vector obtained from exactly these stored coordinates. Therefore it is the component of x that lies in the subspace S , while directions orthogonal to S are discarded.

Step 5 [Bonus] – Concrete example in dimension 5

Let $d = 5$ and $k = 2$. Define $u_1 = \frac{1}{\sqrt{2}}(1, 1, 0, 0, 0)$ and $u_2 = \frac{1}{\sqrt{2}}(0, 0, 1, 1, 0)$.

19. Verify that u_1 and u_2 are orthonormal.

Answer. We have $u_1^\top u_1 = \frac{1}{2}(1^2 + 1^2) = 1$ and $u_2^\top u_2 = \frac{1}{2}(1^2 + 1^2) = 1$. Also $u_1^\top u_2 = 0$, because they have nonzero entries in different coordinates. Thus u_1 and u_2 are orthonormal.

20. Let $x = (2, 4, 1, 3, 10)$. Compute $E(x)$.

Answer. We have $u_1^\top x = \frac{1}{\sqrt{2}}(2 + 4) = \frac{6}{\sqrt{2}} = 3\sqrt{2}$ and $u_2^\top x = \frac{1}{\sqrt{2}}(1 + 3) = \frac{4}{\sqrt{2}} = 2\sqrt{2}$. Thus $E(x) = (3\sqrt{2}, 2\sqrt{2})$.

21. Compute $\hat{x} = D(E(x))$.

Answer. We get $\hat{x} = 3\sqrt{2}u_1 + 2\sqrt{2}u_2$. Since $\sqrt{2}u_1 = (1, 1, 0, 0, 0)$ and $\sqrt{2}u_2 = (0, 0, 1, 1, 0)$, this gives $\hat{x} = 3(1, 1, 0, 0, 0) + 2(0, 0, 1, 1, 0) = (3, 3, 2, 2, 0)$.

22. Which information about x is completely lost by the encoder?

Answer. The encoder keeps the sums $x_1 + x_2$ and $x_3 + x_4$, up to the factor $1/\sqrt{2}$. It loses the difference between x_1 and x_2 , the difference between x_3 and x_4 , and the fifth coordinate x_5 .

23. Determine $\ker(E)$ explicitly in this example.

Answer. A vector $h = (h_1, h_2, h_3, h_4, h_5)$ belongs to $\ker(E)$ iff $h_1 + h_2 = 0$ and $h_3 + h_4 = 0$. Therefore $\ker(E) = \{(\alpha, -\alpha, \beta, -\beta, \gamma) \mid \alpha, \beta, \gamma \in \mathbb{R}\}$. Equivalently, $\ker(E) = \text{span}((1, -1, 0, 0, 0), (0, 0, 1, -1, 0), (0, 0, 0, 0, 1))$.

24. Check that the rank-nullity theorem holds.

Answer. Here $E : \mathbb{R}^5 \rightarrow \mathbb{R}^2$ is surjective, so $\text{rank}(E) = 2$. We found $\dim(\ker(E)) = 3$. Thus $\text{rank}(E) + \dim(\ker(E)) = 2 + 3 = 5 = \dim(\mathbb{R}^5)$, as expected.

Step 6 [Bonus] – Link with PCA

The previous construction becomes a simplified version of PCA when the directions $u_1, \dots, u_k$ are chosen from the data so that they capture the most important directions of variation. PCA chooses a low-dimensional subspace S so that projecting data onto S keeps as much variation as possible.

25. In the previous construction, what is the low-dimensional subspace?

Answer. The low-dimensional subspace is $S = \text{span}(u_1, \dots, u_k)$.

26. What is the compressed representation of x ?

Answer. The compressed representation is $z = E(x) = (u_1^\top x, \dots, u_k^\top x) \in \mathbb{R}^k$.

27. What is the reconstructed approximation of x ?

Answer. The reconstructed approximation is $\hat{x} = D(E(x)) = \sum_{j=1}^k (u_j^\top x) u_j$.

28. Explain why this procedure is linear dimensionality reduction.

Answer. The encoder $E : \mathbb{R}^d \rightarrow \mathbb{R}^k$ is a linear map, and the decoder $D : \mathbb{R}^k \rightarrow \mathbb{R}^d$ is also a linear map. The data is represented using k coordinates instead of d , with $k \ll d$. Hence this is linear dimensionality reduction.

29. Suppose $d = 10\,000$ and $k = 100$. How many coordinates are kept? What fraction of the original number of coordinates is this?

Answer. The representation keeps 100 coordinates instead of 10 000. The fraction is $100/10\,000 = 0.01$, i.e. 1% of the original number of coordinates.

30. Explain why a low-dimensional representation can be useful for image retrieval or face recognition.

Answer. Low-dimensional representations are easier to store, compare, and process. In image retrieval, one can compare embeddings instead of raw images. In face recognition, a low-dimensional vector may keep important information such as identity, while discarding noise or irrelevant pixel-level variation.

Step 7 [Bonus] – Deep image embeddings

In modern AI systems, image embeddings are often produced by neural networks. Such embeddings are usually nonlinear. Nevertheless, the same vector-space ideas remain important.

31. In an image retrieval system, one stores embeddings $z^{(1)}, \dots, z^{(m)} \in \mathbb{R}^k$ for a database of images. Given a query image x_q , one computes $z_q = E(x_q)$. Explain how one could use these embeddings to retrieve visually similar images.

Answer. One can compare z_q with the stored embeddings $z^{(1)}, \dots, z^{(m)}$ using a distance, for example the Euclidean distance $\|z_q - z^{(i)}\|_2$. The retrieved images are the ones whose embeddings are closest to z_q . Thus image retrieval becomes a nearest-neighbor search in the embedding space $\mathbb{R}^k$.

32. What is the main mathematical difference between the linear encoder E studied here and a neural-network image encoder?

Answer. The encoder E studied here is linear: it satisfies $E(\alpha x + \beta y) = \alpha E(x) + \beta E(y)$. A neural-network image encoder is generally nonlinear because it contains nonlinear activation functions and other nonlinear operations. Therefore it does not usually satisfy linearity.

33. Why is it still useful to understand the linear case before studying nonlinear embeddings?

Answer. The linear case introduces the main geometric ideas in a simpler setting: vector spaces, subspaces, coordinates, kernels, ranges, rank, dimension reduction, and reconstruction. These ideas remain important in nonlinear representation learning, even if the maps themselves are more complicated.